

Hijack Execution Flow: DLL

Adversaries may abuse dynamic-link library files (DLLs) in order to achieve persistence, escalate privileges, and evade defenses. DLLs are libraries that contain code and data that can be simultaneously utilized by multiple programs. While DLLs are not malicious by nature, they can be abused through mechanisms such as side-loading, hijacking search order, and phantom DLL hijacking.^[1]

Specific ways DLLs are abused by adversaries include:

DLL Sideloading

Adversaries may execute their own malicious payloads by side-loading DLLs. Side-loading involves hijacking which DLL a program loads by planting and then invoking a legitimate application that executes their payload(s).

Side-loading positions both the victim application and malicious payload(s) alongside each other. Adversaries likely use side-loading as a means of masking actions they perform under a legitimate, trusted, and potentially elevated system or software process. Benign executables used to side-load payloads may not be flagged during delivery and/or execution. Adversary payloads may also be encrypted/packed or otherwise obfuscated until loaded into the memory of the trusted process.

Adversaries may also side-load other packages, such as BPLs (Borland Package Library).^[2]

Adversaries may chain DLL sideloading multiple times to fragment functionality hindering analysis. Adversaries using multiple DLL files can split the loader functions across different DLLs, with a main DLL loading the separated export functions.^[3] Spreading loader functions across multiple DLLs makes analysis harder, since all files must be collected to fully understand the malware's behavior. Another method implements a "loader-for-a-loader", where a malicious DLL's sole role is to load a second DLL (or a chain of DLLs) that contain the real payload.^[4]

DLL Search Order Hijacking

Adversaries may execute their own malicious payloads by hijacking the search order that Windows uses to load DLLs. This search order is a sequence of special and standard search locations that a program checks when loading a DLL. An adversary can plant a trojan DLL in a directory that will be prioritized by the DLL search order over the location of a legitimate library. This will cause Windows to load the malicious DLL when it is called for by the victim program.^[1]

DLL Redirection

Adversaries may directly modify the search order via DLL redirection, which after being enabled (in the Registry or via the creation of a redirection file) may cause a program to load a DLL from a different location.^{[5][6]}

Phantom DLL Hijacking

Adversaries may leverage phantom DLL hijacking by targeting references to non-existent DLL files. They may be able to load their own malicious DLL by planting it with the correct name in the location of the missing module.^{[7][8]}

DLL Substitution

Adversaries may target existing, valid DLL files and substitute them with their own malicious DLLs, planting them with the same name and in the same location as the valid DLL file.^[9]

Programs that fall victim to DLL hijacking may appear to behave normally because malicious DLLs may be configured to also load the legitimate DLLs they were meant to replace, evading defenses.

Remote DLL hijacking can occur when a program sets its current directory to a remote location, such as a Web share, before loading a DLL.^{[10][11]}

If a valid DLL is configured to run at a higher privilege level, then the adversary-controlled DLL that is loaded will also be executed at the higher level. In this case, the technique could be used for privilege escalation.

ID: T1574.001

Sub-technique of: [T1574](#)



Tactics: [Stealth](#), [Execution](#)



Platforms: Windows

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Procedure Examples

ID	Name	Description
C0057	3CX Supply Chain Attack	During the 3CX Supply Chain Attack , AppleJeus splits functionally across multiple .dll files using export functions, such as DLLGetClassObject, to execute code from an embedded .dll file within another .dll file. AppleJeus has also used DLL search order hijacking via the IKEEXT service, running with LocalSystem privileges, to load the TAXHAUL DLL for persistence. ^{[12][13]}
S9027	ANELLDR	ANELLDR can use DLL sideloading from a legitimate application to initiate execution. ^[14]
G0099	APT-C-36	APT-C-36 has used side-loading to execute the HijackLoader payload. ^[15]
G0073	APT19	APT19 launched an HTTP malware variant and a Port 22 malware variant using a legitimate executable that loaded the malicious DLL. ^[16]
G0022	APT3	APT3 has been known to side load DLLs with a valid version of Chrome with one of their tools.
G0050	APT32	APT32 ran legitimately-signed executables from Symantec and McAfee which load a malicious DLL. The group also side-loads its backdoor by dropping a library and a legitimate, signed executable (AcroTranscoder). ^{[19][20][21]}
G0096	APT41	APT41 has used search order hijacking to execute malicious payloads, such as Winnti for Windows . ^[22] APT41 has also used legitimate executables to perform DLL side-loading of their malware. ^[23]
C0040	APT41 DUST	APT41 DUST involved the use of DLL search order hijacking to execute DUSTTRAP . ^[24] APT41 DUST used also DLL side-loading to execute DUSTTRAP via an AhnLab uninstaller.
G0143	Aquatic Panda	Aquatic Panda has used DLL search-order hijacking to load <code>exe</code> , <code>dll</code> , and <code>dat</code> files into memory. ^[25] Aquatic Panda loaded a malicious DLL into the legitimate Windows Security Health Service executable (<code>SecurityHealthService.exe</code>) to execute malicious code on victim systems. ^[26]
S9031	AshTag	AshTag has enabled execution via DLL sideloading using a legitimate executable paired with a malicious DLL named wtsapi32. ^[27]
S0373	Astaroth	Astaroth can launch itself via DLL Search Order Hijacking. ^[28]
G0135	BackdoorDiplomacy	BackdoorDiplomacy has executed DLL search order hijacking. ^[29]
S0128	BADNEWS	BADNEWS typically loads its DLL file into a legitimate signed Java or VMware executable. ^[30]
S0127	BBSRAT	DLL side-loading has been used to execute BBSRAT through a legitimate Citrix executable, ssonsvr.exe. The Citrix executable was dropped along with BBSRAT by the dropper. ^[32]
G0098	BlackTech	BlackTech has used DLL side loading by giving DLLs hardcoded names and placing them in searched directories. ^[33]
S1226	BOOKWORM	BOOKWORM has used DLL side-loading to execute the malicious payload. ^{[34][35]} BOOKWORM has also side-loaded DLL components into a legitimate process, including Microsoft Malware Protection <code>MsMpEng.exe</code> and Kaspersky Anti-Virus <code>ushata.exe</code> . ^[36]
S0415	BOOSTWRITE	BOOSTWRITE has exploited the loading of the legitimate Dwrite.dll file by actually loading the gdi library, which then loads the gdiplus library and ultimately loads the local Dwrite dll. ^[37]
G0060	BRONZE BUTLER	BRONZE BUTLER has used legitimate applications to side-load malicious DLLs. ^[38]
S1063	Brute Ratel C4	Brute Ratel C4 has used search order hijacking to load a malicious payload DLL as a dependency to a benign application packaged in the same ISO. ^[39] Brute Ratel C4 has loaded a malicious DLL by spoofing the name of the legitimate Version.DLL and placing it in the same folder as the digitally-signed Microsoft binary OneDriveUpdater.exe. ^[39]
S1237	CANONSTAGER	CANONSTAGER has abused legitimate executables to side-load malicious DLLs. ^[40]
S0631	Chaes	Chaes has used search order hijacking to load a malicious DLL. ^[41]
G0114	Chimera	Chimera has used side loading to place malicious DLLs in memory. ^[42]

ID	Name	Description
S1041	Chinoxy	Chinoxy can use a digitally signed binary ("Logitech Bluetooth Wizard Host Process") to load its dll into memory. ^[43]
G1021	Cinnamon Tempest	Cinnamon Tempest has used search order hijacking to launch Cobalt Strike Beacons. ^{[44][45]} Cinnamon Tempest has also abused legitimate executables to side-load weaponized DLLs. ^[46]
S1236	CLAIMLOADER	CLAIMLOADER has used a legitimately signed executable to execute a malicious payload within a DLL file. ^[47]
S0660	Clambling	Clambling can store a file named <code>mpsvc.dll</code> , which opens a malicious <code>mpsvc.mui</code> file, in the same folder as the legitimate Microsoft executable <code>MsMpEng.exe</code> to gain execution. ^{[48][49]}
S1235	CorKLOG	CorKLOG has leveraged legitimate binaries to conduct DLL side-loading. ^[50]
S0538	Crutch	Crutch can persist via DLL search order hijacking on Google Chrome, Mozilla Firefox, or Microsoft OneDrive. ^[51]
G1034	Daggerfly	Daggerfly has used legitimate software to side-load PlugX loaders onto victim systems. ^[52] Daggerfly is also linked to multiple other instances of side-loading for initial loading activity. ^[53]
S1111	DarkGate	DarkGate includes one infection vector that leverages a malicious "KeyScramblerE.DLL" library that will load during the execution of the legitimate KeyScrambler application. ^[54]
S0354	Denis	Denis exploits a security vulnerability to load a fake DLL and execute its code. ^[19]
S0134	Downdelph	Downdelph uses search order hijacking of the Windows executable sysprep.exe to escalate privileges. ^[55]
S0384	Dridex	Dridex can abuse legitimate Windows executables to side-load malicious DLL files. ^[56]
G1006	Earth Lusca	Earth Lusca has placed a malicious payload in <code>%WINDIR%\SYSTEM32\oci.dll</code> so it would be sideloaded by the MSDTC service. ^[57]
S0624	Ecipekac	Ecipekac can abuse the legitimate application policytool.exe to load a malicious DLL. ^[58]
S0554	Egregor	Egregor has used DLL side-loading to execute its payload. ^[59]
S0363	Empire	Empire contains modules that can discover and exploit various DLL hijacking opportunities. ^[60]
G0120	Evilnum	Evilnum has used the malware variant, TerraTV, to load a malicious DLL placed in the TeamViewer directory, instead of the original Windows DLL located in a system folder. ^[61]
G1016	FIN13	FIN13 has used IISCrack.dll as a side-loading technique to load a malicious version of httpodbc.dll on old IIS Servers (CVE-2001-0507). ^[62]
S0182	FinFisher	FinFisher uses DLL side-loading to load malicious programs. ^{[63][64]} A FinFisher variant also uses DLL search order hijacking. ^{[63][65]}
S0661	FoggyWeb	FoggyWeb 's loader has used DLL Search Order Hijacking to load malicious code instead of the legitimate <code>version.dll</code> during the <code>Microsoft.IdentityServer.ServiceHost.exe</code> execution process. ^[66]
G0093	GALLIUM	GALLIUM used DLL side-loading to covertly load PoisonIvy into memory on the victim machine. ^[67]
S0032	gh0st RAT	A gh0st RAT variant has used DLL side-loading. ^[68]
S0477	Goopy	Goopy has the ability to side-load malicious DLLs with legitimate applications from Kaspersky, Microsoft, and Google. ^[20]
S1229	Havoc	Havoc has leveraged legitimate executables to side-load malicious payloads. ^[69]
G0126	Higaisa	Higaisa 's JavaScript file used a legitimate Microsoft Office 2007 package to side-load the <code>OINFO12.OCX</code> dynamic link library. ^[70]
S0009	Hikit	Hikit has used DLL to load <code>oci.dll</code> as a persistence mechanism. ^[71]

ID	Name	Description
S1230	HIUPAN	HIUPAN has abused legitimate executables to side-load malicious DLLs to include the legitimate exe UsbConfig.exe. ^{[72][73]}
S0070	HTTPBrowser	HTTPBrowser abuses the Windows DLL load order by using a legitimate Symantec anti-virus binary, VPDN_LU.exe, to load a malicious DLL that mimics a legitimate Symantec DLL, navlu.dll. ^[74] HTTPBrowser has also used DLL side-loading. ^[75]
S1097	HUI Loader	HUI Loader can be deployed to targeted systems via legitimate programs that are vulnerable to DLL search order hijacking. ^[45]
S0398	HyperBro	HyperBro has used a legitimate application to sideload a DLL to decrypt, decompress, and run a payload. ^{[76][77]}
S0260	InvisiMole	InvisiMole can be launched by using DLL search order hijacking in which the wrapper DLL is placed in the same folder as explorer.exe and loaded during startup into the Windows Explorer process instead of the legitimate library. ^[78]
S9029	IronWind	IronWind has used DLL sideloading for execution. ^[69]
S0528	Javali	Javali can use DLL side-loading to load malicious DLLs into legitimate executables. ^[28]
S0585	Kertdown	Kertdown can use DLL side-loading to load malicious DLLs. ^[79]
G0032	Lazarus Group	Lazarus Group has replaced <code>win_fw.dll</code> , an internal component that is executed during IDA Pro installation, with a malicious DLL to download and execute a payload. ^[80] Lazarus Group utilized DLL side-loading to execute malicious payloads through abuse of the legitimate processes <code>wsmprovhost.exe</code> and <code>dfrgui.exe</code> . ^[81]
S9020	LODEINFO	LODEINFO can use legitimate EXE files to sideload malicious DLLs. ^[82]
S1101	LoFiSe	LoFiSe has been executed as a file named DsNcDiag.dll through side-loading. ^[83]
S0582	LookBack	LookBack side loads its communications module as a DLL into the <code>libcurl.dll</code> loader. ^[84]
G1014	LuminousMoth	LuminousMoth has used legitimate executables such as <code>winword.exe</code> and <code>igfxem.exe</code> to side-load their malware. ^{[85][86]}
S1213	Lumma Stealer	Lumma Stealer has leveraged legitimate applications to then side-load malicious DLLs during execution. ^[87]
S0530	Melcoz	Melcoz can use DLL hijacking to bypass security controls. ^[28]
G0045	menuPass	menuPass has used DLL side-loading to launch versions of Mimikatz and PwDump6 as well as UPPERCUT . ^{[88][89][90]} menuPass has also used DLL search order hijacking. ^[91]
S1059	metaMain	metaMain can support an HKCMD sideloading start method. ^[92]
S0455	Metamorfo	Metamorfo has side-loaded its malicious DLL file. ^{[93][94][95]}
S0280	MirageFox	MirageFox is likely loaded via DLL hijacking into a legitimate McAfee binary. ^[96]
G1054	MirrorFace	MirrorFace has used legitimate EXE files to load malicious DLLs via sideloading. ^{[82][97][98][99]}
G0069	MuddyWater	MuddyWater maintains persistence on victim networks through side-loading dlls to trick legitimate programs into running malware. ^[100]
G0129	Mustang Panda	Mustang Panda has used a legitimately signed executable to execute a malicious payload within a DLL file. ^{[101][102][103][34][104][105][106][107][108][109][36][110][50][111]} Mustang Panda has abused legitimate executables to side-load malicious DLLs. ^{[112][113][47][72][40]}
G0019	Naikon	Naikon has used DLL side-loading to load malicious DLL's into legitimate executables. ^[114]
S0630	Nebulae	Nebulae can use DLL side-loading to gain execution. ^[115]
S1100	Ninja	Ninja loaders can be side-loaded with legitimate and signed executables including the VLC.exe media player. ^[83]
S9025	NOOPLDR	NOOPLDR can be executed via sideloading. ^{[99][116]}

ID	Name	Description
C0012	Operation CuckooBees	During Operation CuckooBees , the threat actors used the legitimate Windows services <code>IKEEXT</code> and <code>PrintNotify</code> to side-load malicious DLLs. ^[117]
S1233	PAKLOG	PAKLOG has leveraged legitimate binaries to conduct DLL side-loading. ^[50]
S0664	Pandora	Pandora can use DLL side-loading to execute malicious payloads. ^[77]
G0040	Patchwork	A Patchwork .dll that contains BADNEWS is loaded and executed using DLL side-loading. ^[118]
S1102	Pcexte	Pcexte has been distributed and executed as a DLL file named Vspmsg.dll via DLL side-loading. ^[83]
S0013	PlugX	PlugX has the ability to use DLL search order hijacking for installation on targeted systems. ^[119] ^[110] PlugX has also used DLL side-loading to evade anti-virus. ^{[18][75][120][88][121][48][122]} PlugX has also used a legitimately signed executable to side-load a malicious payload within a DLL file. ^{[101][103][104][110][123]}
S0194	PowerSploit	PowerSploit contains a collection of Privesc-PowerUp modules that can discover and exploit DLL hijacking opportunities in services and processes. ^{[124][125]}
S1046	PowGoop	PowGoop can side-load <code>Goopdate.dll</code> into <code>GoogleUpdate.exe</code> . ^{[100][126]}
S0113	Prikormka	Prikormka uses DLL search order hijacking for persistence by saving itself as ntshrui.dll to the Windows directory so it will load before the legitimate ntshrui.dll saved in the System32 subdirectory. ^[127]
S1228	PUBLOAD	PUBLOAD has abused legitimate executables to side-load malicious DLLs. ^{[103][112][113][72][108]}
S0650	QakBot	QakBot has the ability to use DLL side-loading for execution. ^[129]
S0629	RainyDay	RainyDay can use side-loading to run malicious executables. ^[115]
S0458	Ramsay	Ramsay can hijack outdated Windows application dependencies with malicious versions of its own DLL payload. ^[130]
S1130	Raspberry Robin	Raspberry Robin can use legitimate, signed EXE files paired with malicious DLL files to load and run malicious payloads while bypassing defenses. ^[131]
S0662	RCSession	RCSession can be installed via DLL side-loading. ^{[132][48][122]}
C0047	RedDelta Modified PlugX Infection Chain Operations	Mustang Panda used DLL search order hijacking on vulnerable applications to install PlugX payloads during RedDelta Modified PlugX Infection Chain Operations . ^[133]
S0153	RedLeaves	RedLeaves is launched through use of DLL search order hijacking to load a malicious dll. ^[134]
S9026	ROAMINGHOUSE	ROAMINGHOUSE can use a legitimate EXE to sideload a malicious DLL named JSFC.dll. ^[135] ROAMINGHOUSE has also used ScnCf32.exe to sideload vsodscpl.dll to enable UPPERCUT execution. ^[14]
G0048	RTM	RTM has used search order hijacking to force TeamViewer to load a malicious DLL. ^[136]
S0074	Sakula	Sakula uses DLL side-loading, typically using a digitally signed sample of Kaspersky Anti-Virus (AV) 6.0 for Windows Workstations or McAfee's Outlook Scan About Box to load malicious DLL files. ^[137]
G1008	SideCopy	SideCopy has used a malicious loader DLL file to execute the <code>credwiz.exe</code> process and side-load the malicious payload <code>Duser.dll</code> . ^[138]
G0121	Sidewinder	Sidewinder has used DLL side-loading to drop and execute malicious payloads including the hijacking of the legitimate Windows application file rekeywiz.exe. ^[139]
S1232	SplatDropper	SplatDropper has leveraged legitimate binaries to conduct DLL side-loading. ^[50]
S1227	StarProxy	StarProxy has been side-loaded by the legitimate, signed executable, IsoBurner.exe. ^[111]
G1046	Storm-1811	Storm-1811 has deployed a malicious DLL (7z.DLL) that is sideloaded by a modified, legitimate installer (7zG.exe) when that installer is executed with an additional command line parameter of <code>b</code> at runtime to load a Cobalt Strike beacon payload. ^[140]

ID	Name	Description
S1183	StrelaStealer	StrelaStealer has sideloaded a DLL payload using a renamed, legitimate <code>msinfo32.exe</code> executable. ^[141]
S0663	SysUpdate	SysUpdate can load DLLs through vulnerable legitimate executables. ^[77]
S0098	T9000	During the T9000 installation process, it drops a copy of the legitimate Microsoft binary <code>igfxtray.exe</code> . The executable contains a side-loading weakness which is used to load a portion of the malware. ^[142]
G0027	Threat Group-3390	Threat Group-3390 has performed DLL search order hijacking to execute their payload. ^[143] Threat Group-3390 has also used DLL side-loading, including by using legitimate Kaspersky antivirus variants as well as <code>rc.exe</code> , a legitimate Microsoft Resource Compiler. ^{[75][144][145][76]}
S1239	TONESHELL	TONESHELL has abused legitimate executables to side-load malicious DLLs. ^{[112][147][106][107][108][148]} TONESHELL has also been loaded via DLL side-loading, using legitimate, signed executables to include: <code>FastVD.exe</code> , <code>Bandizip.exe</code> and <code>gpgconf.exe</code> . ^[111]
G0131	Tonto Team	Tonto Team abuses a legitimate and signed Microsoft executable to launch a malicious DLL.
G0081	Tropic Trooper	Tropic Trooper has been known to side-load DLLs using a valid version of a Windows Address Book and Windows Defender executable with one of their tools. ^{[150][151]}
S0275	UPPERCUT	UPPERCUT has been sideloaded through a legitimately signed application from the JustSystems Corporation. ^[152]
G1047	Velvet Ant	Velvet Ant has used malicious DLLs executed via legitimate EXE files through DLL search order hijacking to launch follow-on payloads such as PlugX . ^[153]
S0612	WastedLocker	WastedLocker has performed DLL hijacking before execution. ^[154]
S0579	Waterbear	Waterbear has used DLL side loading to import and load a malicious DLL loader. ^[33]
S0109	WEBC2	Variants of WEBC2 achieve persistence by using DLL search order hijacking, usually by copying the DLL file to <code>%SYSTEMROOT% (C:\WINDOWS\ntshrui.dll)</code> . ^[155]
G0107	Whitefly	Whitefly has used search order hijacking to run the loader <code>Vcrodut</code> . ^[156]
S0176	Wingbird	Wingbird side loads a malicious file, <code>sspisrv.dll</code> , in part of a spoofed <code>Issas.exe</code> service. ^{[157][158]}
G0090	WIRTE	WIRTE has used RAR archives containing a legitimate executable and a lure document to execute malicious DLLs via sideloading. ^[69]
S0230	ZeroT	ZeroT has used DLL side-loading to load malicious payloads. ^{[159][160]}

Mitigations

ID	Mitigation	Description
M1013	Application Developer Guidance	When possible, include hash values in manifest files to help prevent side-loading of malicious libraries.
M1047	Audit	Use auditing tools capable of detecting DLL search order hijacking opportunities on systems within an enterprise and correct them. Toolkits like the PowerSploit framework contain PowerUp modules that can be used to explore systems for DLL hijacking weaknesses.^[161] Use the program <code>sxstrace.exe</code> that is included with Windows, along with manual inspection, to check manifest files for side-by-side problems in software.^[162]
M1038	Execution Prevention	Identify and block potentially malicious software executed through DLL hijacking by using application control solutions capable of blocking DLLs loaded by legitimate software. ^[163]
M1044	Restrict Library Loading	Disallow loading of remote DLLs. This is included by default in Windows Server 2012+ and is available by patch for XP+ and Server 2003+.^[164] Enable Safe DLL Search Mode to move the user's current folder later in the search order. This is included by default in modern versions of Windows; the associated Windows Registry key is located at <code>HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\SafeDLLSearchMode</code>.^[165]
M1051	Update Software	Update software regularly to include patches that fix DLL side-loading vulnerabilities.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0201	Detection Strategy for Hijack Execution Flow for DLLs	AN0577	DLL hijacking behaviors including unexpected DLL loads from non-standard directories, replacement of DLLs, phantom DLL insertion, redirection file creation, and substitution of legitimate DLLs. Defender correlates file system modifications, registry changes, and module load telemetry to detect abnormal DLL behavior in trusted processes.

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